CLA 2026
Crash Course

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Statistical Methods for Linguistic Data

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Crash Course Roadmap

1 — Why stats matter

2 — Understanding your data

3 — The pipeline

4 — Model families

5 — Common Pitfalls

🚀 — Practical Component

Goals of the crash course:

1. Cover the full pipeline from identifying your data type to interpreting model output (A LOT in 2 hours, but it is a crash course after all!)
2. Build enough confidence to get started on your own analyses
3. Develop the foundation to learn further; help get a better sense of what to search for
4. Leave with reference guides (these slides, the code notebook) you can return to

All code runs in Google Colab. No local installation needed. You can follow along on your machine or just watch.

01

Why should we want to understand statistics?

The Cost of Getting It Wrong

Wrong conclusions

A mismatch between data type and model can produce biased estimates, even when the code runs fine and no error is thrown.

Failed peer review

Reviewers increasingly flag pseudoreplication, missing methods, and inappropriate model specifications.

Irreproducibility

Analyses that cannot be reproduced (by others or by you six months later) are a problem. Statistical transparency is part of the fix.

✓ The good news: a principled workflow and good documentation handles a lot of the difficulty.

Some Best Practices

Predetermine structure

Specify your dependent variable, independent variables, and analysis methods before collecting data.

Visualize first

Always plot your data before fitting a model. Summary statistics can be deeply misleading (see Anscombe's Quartet).

Report in full

Include model specification, random effects structure, software and package versions, etc.; report R version, not R Studio version.

Share your data and code

Open Science Framework (OSF), GitHub, or supplementary materials. Makes your work verifiable and reproducible.

02

Understanding Your Data

Variable Types

Continuous	F0, voice onset time, reaction time, duration, formant values, reading times, gaze duration
Binary	Correct/incorrect, target/non-target, yes/no judgments, re-reading yes/no
Count	Corpus token frequency, distributional data, error counts, number of fixations
Nominal	Language or dialect, experimental condition, speaker identity
Ordinal	Likert scale (1–5, 1–7), proficiency ratings, complexity scores

Your outcome variable type determines your model family. Everything else follows from this.

What Your Data Looks Like

partID	item	condition	rating	rt_ms	accuracy
P01	it01	subject-gap	6	423.7	1
P01	it02	object-gap	3	681.3	0
P02	it01	subject-gap	7	398.5	1
P02	it02	object-gap	2	712.2	0
P03	it01	subject-gap	5	501.9	1
...	...	...	...	...	...
nominal	nominal	nominal	ordinal	continuous	binary

each row = one observation

columns = variables

'tidy' format (what most stats programs expect)

Long vs. Wide Data

Most analyses need long format data, but some data collection is in wide format.

Wide format

One row per participant. Conditions spread across columns.

partID	cond_A	cond_B	cond_C	cond_D
P01	423	651	498	712
P02	381	598	445	689
P03	512	703	521	798
…	…	…	…	…

Problems: Model calls cannot read this. Each condition needs its own row. Here, adding a new condition means adding a new column.

Long format

One row per observation. Condition is a column.

partID	condition	cond_B
P01	cond_A	423
P01	cond_B	651
P01	cond_C	498
P01	cond_D	712
…	…	…

Fix: One row per observation maps directly to the model. Easy to add new variables as new columns.

```
Reshape in R with pivot_longer() function from the tidyr package (or tidyverse)
```

Dependent Variable vs. Independent Variables

Dependent Variable (Outcome)

What you are trying to explain or predict

Examples:
- Acceptability rating
- Reaction/response time
- Response accuracy
- Corpus token frequency

This determines your model family.

Independent Variables (Predictors)

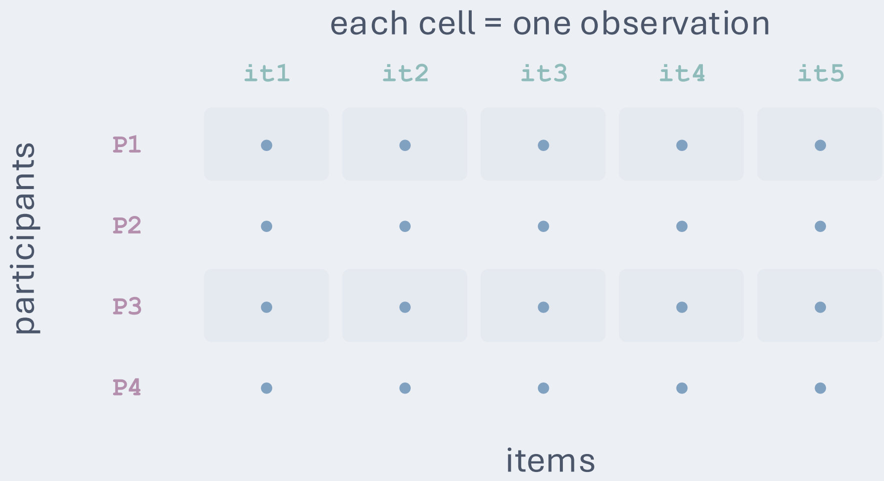
What you think explains the outcome

Examples:
- Syntactic condition
- Language proficiency level
- Speech rate
- Speaker age or gender

These DO NOT determine model family.

The Structure Problem: Non-Independence

**Most linguistic data is NOT a simple random sample.
Observations share structure.**



- P1's responses share something (their language background, attention, etc.)

- it1's responses share something (its difficulty, linguistic properties, etc.)

Treating rows as independent → pseudoreplication → possible inflated false positive rate

Solution: Treat participants AND items as random effects

03

The Analysis Pipeline

The Analysis Pipeline

1. **Visualize**
   Histograms, boxplots, strip charts, and more. Look before you model.

2. **Choose model**
   Match model family to outcome variable type.

3. **Fit and check**
   Residuals, Q-Q plots, convergence warnings.

4. **Interpret**
   Coefficients, confidence intervals, and *p*-values in context.

5. **Report**
   Transparently: model specs, random effects, software versions.

Step 1: Visualize Before (and After) You Model

Plot type	Best for	What it tells you	R Code
Histogram	Continuous / skewed outcomes	Shape of the distribution: skew, bimodality, floor/ceiling effects, outliers.	`geom_histogram()`
Violin + boxplot	Comparing groups	Full distribution shape and spread across conditions. Prefer over bar charts which hide distributional information.	`geom_violin() + geom_boxplot()`
Strip chart	Any numerical outcome; small to medium *n*	Every individual observation. Reveals sample size reality and whether the group pattern holds for individuals.	`geom_jitter() + stat_summary()`
Residual plot	After fitting linear models	Random scatter around zero = assumptions met. Fan shape = heteroscedasticity. Curve = missed nonlinearity.	`plot(model)`
Q-Q plot	After fitting linear models	Points on the diagonal = normally distributed residuals. Heavy tails = outliers or non-normal errors.	`qqnorm(resid(model))`

In the practical component we will generate all five plot types for the study data.

Key Concepts

Mean

The average; add all values up and divide by how many there are.

e.g., five RTs of 600, 650, 700, 750, 800ms; mean = 700ms

Variance

How spread out values are around the mean. High variance = scattered; low variance = clustered tightly.

e.g., two conditions, same mean reaction time, but one is all over the place and one is very consistent; same mean, different variance

Residuals

The gap between what the model predicted and what actually happened, i.e., what the model could not explain.

e.g. model predicts 650ms, participant takes 720ms; residual = 70ms

Fixed effects

The predictors in your model whose relationship with the outcome you want to estimate; includes things you deliberately manipulated and things you measured and want to account for.

e.g., experimental conditions (you designed the experiment around them), covariates such as participant age, gender, or language proficiency

Random effects

Sources of variation you want to control for but are not specifically testing; typically participants and items. Random effects absorb baseline differences so your fixed effects remain uncontaminated.

e.g., P01 is just faster than average at everything; a random effect captures that so it does not distort your results

Step 2: Choosing Your Model

Ask these questions in order:

What type is my outcome variable?	→ Determines model family
Is the data independent, or do observations share structure?	→ If participants/items are repeated, add random effects
Is there overdispersion in count data?	→ If variance > mean, use negative binomial instead of Poisson
Do I have crossed or nested random effects?	→ Crossed: (1|participant) + (1|item) → Nested: (1|school/student)
Did my model converge?	→ If not, simplify random effects structure, or consider another model architecture

Model Selection

Model selection involves three decisions:

1. Model family

Determined by outcome type:

Continuous	`lmer()`
Binary	`glmer()`
Ordinal	`clmm()`
Count	`glmer.nb()`

→

2. Model specification

Fixed effects

Pre-specify all theoretically motivated predictors and covariates before fitting.

Interactions

Add if effect of one predictor depends on another. Always include main effects too.

Random effects

Participant and item, in applicable; accounts for baseline differences.

Random slopes

Add when you want to allow the effect of a predictor to vary across each cluster in your random effect(s).

→

3. Model comparison

Akaike Information Criterion (AIC)

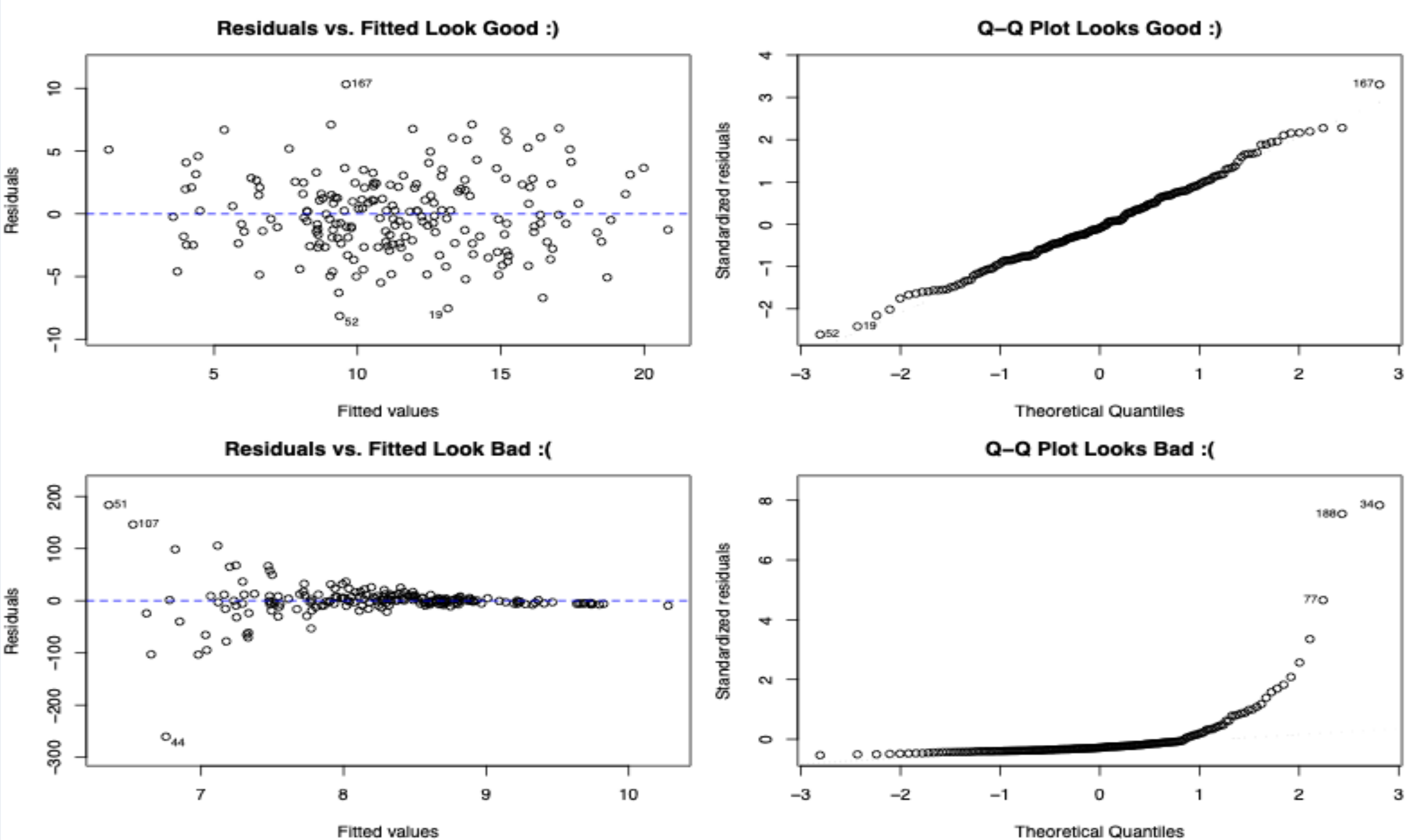
Lower AIC = better fit
Difference > 2 is meaningful
Penalises model complexity
`AIC(m1, m2)`

Likelihood ratio test (LRT)

Tests if adding a term significantly improves fit; models must be nested; use REML = FALSE for LRT
`anova(m_null, m_full)`

A note on model comparison: AIC is used for comparing non-nested models or multiple candidates at once; LRT is used for testing whether a specific term (e.g., a fixed or random effect, an interaction, etc.) significantly improves a simpler model

Step 3: Fit and Check Assumptions



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Model Families for Linguistic Data

What is Regression?

Regression models the relationship between an outcome (what you want to explain) and one or more predictors (what you think explains it), and quantifies how much each predictor matters to the story.

The equation:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Do you remember $y = mx + b$?

- y outcome variable
- β_0 intercept, predicted value of y when x = 0
- β_1 coefficient, how much y changes per unit of x
- x predictor variable
- ε residual, what the model does not explain

The coefficient β_1 is the answer to your research question:
How much does *x* predict *y*, and in which direction?

Why Regression? Beyond t-tests and ANOVAs

t-tests and ANOVAs are not wrong, but regression is more flexible.

Handles non-independence

t-tests and ANOVAs assume independent observations. Participants respond to multiple items; items are seen by multiple participants. Mixed effects models this structure directly.

Generalizes across outcome types

t-tests require continuous, normally distributed outcomes. Regression offers a principled model family for ordinal, binary, and count data.

Handles continuous predictors

ANOVA treats all predictors as categorical. Regression includes word frequency, speech rate, or trial number as covariates naturally.

Effect sizes and uncertainty built in

Regression gives coefficients, confidence intervals, and effect sizes as part of the output; reviewers expect these measures to be reported as they are more informative than $p < .05$.

ANOVAs and t-tests are special cases of the linear models. We are generalizing, not starting over.

Summary of Common Model Families

Outcome type	Model family	R functions	Examples in linguistics
Continuous	Linear regression / Linear mixed-effects (LME)	`lm() / lmer()`	Reaction/response times, VOT, F0, reading duration
Binary	Logistic regression / Generalized Linear Mixed Effects (GLME) logistic regression	`glm(family=binomial) / glmer()`	Correct/incorrect, choice tasks, yes/no judgments
Ordinal	Ordinal regression	`clm() / clmm()`	Likert acceptability ratings (ordered, not numeric), proficiency ratings
Count	Poisson / Negative binomial	`glm(family=poisson) / glm.nb()`	Corpus frequencies, error counts, token rates
Nominal	Multinomial regression	`multinom() / mlogit()`	Choice between 3+ unordered categories, e.g., language or dialect; classification tasks

Model Assumptions

Shared across all models

Correct model specification
Right predictors included; relationship in the right functional form

Independence of residuals
After accounting for random effects, residuals should not be systematically related

No extreme multicollinearity
Predictors should not be so correlated the model cannot separate their effects

Sufficient sample size
Enough observations relative to the number of parameters being estimated

Things to check after fitting

Residuals vs. fitted
Random scatter around zero checks for linearity and homoscedasticity (constant variance)

Q-Q plot of residuals
Points on the diagonal checks normality of residuals for linear models

Convergence check
No warnings from the optimizer; if the model did not converge, simplify the random effects structure

Dispersion check
For count models only; variance/mean ratio should be close to 1 for Poisson regression

Linear Mixed Effects: Reading the Output

```
lmer(RT ~ condition + (1|participant) + (1|item), data = your_data)
```

```
Fixed effects:
               Estimate  Std.Error t value
(Intercept)    523.4     18.2      28.76
conditionObj   87.3      14.6      5.98

Random effects:
 Groups       Variance  Std.Dev.
 participant  2341.2    48.4
 item         891.4     29.9
 Residual     3102.1    55.7
```

Intercept (523.4 ms)
Mean RT for the reference condition (subject-gap)

conditionObj (+87.3 ms)
Object-gap sentences take 87.3 ms longer on average. This is the effect of interest.

Random effects variance
Variance due to participants (48ms SD) and items (30ms SD) is separated from residual noise.

t value ≈ z for large n. No p-values by default in lmer; use lmerTest package to get p-values.

Report: β = 87.3ms, *SE* = 14.6, *t*(df) = 5.98, *p* < .001, along with confidence interval calculated afterward.

Logistic Regression: Reading the Output

```
glmer(correct ~ condition + (1|participant) + (1|item), family = binomial,
      data = df)
```

```
Fixed effects:

               Estimate  Std.Error  z value  Pr(>|z|)
(Intercept)     1.386      0.241     5.75    <.001
conditionObj   -0.916      0.198    -4.63    <.001
```

Log-odds (logit) scale
Coefficients are not probabilities. A positive value means higher probability of the outcome (coded 1).

Converting to probability
plogis() in R converts log-odds → probability
exp() converts log-odds → odds ratio

Odds ratio for conditionObj
exponentiate the coefficient to get an odds ratio
exp(−0.916) = 0.40 → object-gap responses have 40% of the odds (60% lower odds) of being correct

In a logistic regression model, the intercept represents the log-odds of success when all categorical predictors are at their reference level, and all continuous predictors are at zero.

To interpret: plogis(1.386) = 0.80 (predicted probability of 80% correct response on subject-gap)
plogis(1.386 + (−0.916)) = 0.62 (predicted probability of 62% correct response on object-gap)

Ordinal Regression: Rating Scales Spotlight

Common mistake

Treating 1–7 Likert ratings as continuous and running linear regression

Why it is incorrect:
- "4" ≠ twice as acceptable as "2"
- Distances between levels aren't equal
- Can predict impossible values (0.3, 8.1)
- Violates distributional assumptions

Also: reporting means hides distributional shape; two conditions can have identical means but very different distributions

Better approach

Ordinal regression with clmm()

What it does:
- Models the ordered categories directly
- Estimates thresholds between each level
- Handles random effects properly

In R:

```
library(ordinal)
clmm(rating ~ condition +
   (1|participant) + (1|item),
  data = df)
```

Reading output from ordinal regression works just like logistic regression: coefficients are on the log-odds scale, exponentiate for odds ratios, interpret direction and significance the same way.

Count Models: Corpus and Frequency Data

When to use Poisson

✓ Outcome is a non-negative integer count

✓ Zero is a possible and meaningful value

✓ Variance ≈ Mean (equidispersion)

Examples:
- Token frequency per million words
- Number of disfluencies or errors per utterance
- Error count per participant

R: `glm(count ~ predictor, family = poisson)`

When to use Negative Binomial

Same as Poisson, but Variance > Mean

Overdispersion: more spread than Poisson allows

Very common in corpus data

How to check:
i. Fit Poisson, run dispersiontest() from AER
ii. Or compare AIC: lower = better fit

R: `glm.nb(count ~ predictor)`

Never use linear regression on count data as it can predict negative counts and it mishandles the mean-variance relationship.

Poisson and Negative Binomial: Reading the Output

```
glmer(count ~ predictor1 + offset(log(predictor2)) + (1|random effect), family = poisson, data = your_data)
```

Fixed effects:

	Estimate	Std.Error	z value	Pr(>|z|)
(Intercept)	-2.341	0.142	-16.49	<.001
corpus_typenews	-0.612	0.098	-6.24	<.001
corpus_typesocial	0.238	0.091	2.62	.009

Coefficients are log rate ratios; exponentiate for interpretation

exp(− 0.612) = 0.54, where (0.54 − 1)*100 = −46
News documents have 46% lower subject gap rate than magazines (reference)

exp(0.541) = 1.72, where (1.72 − 1)*100 = 72
Social media has 72% higher subject gap rate than magazines (reference)

Poisson assumption: variance = mean. Check with dispersiontest() from AER package. If violated, switch to glmer.nb() for negative binomial.

Negative binomial: adds a dispersion parameter to model variance > mean. Use glmer.nb(); same syntax as above, no family argument needed.

S I D E   Q U E S T

A Different Philosophy

Frequentist vs. Bayesian: Two Philosophies

Frequentist

What it asks:
If H_0 were true, how likely is data this extreme?

Parameters are fixed (unknown) constants and data are the random quantity, i.e., if you ran your study again, you would get different data.

Output:
p-values, confidence intervals, test statistics

✓ "We reject H_0"
✗ "Probability that H_0 is true"

Bayesian

What it asks:
Given the data, what should we believe about the parameter?

Parameters have probability distributions
Prior belief updated by evidence → posterior

Output:
Posterior distributions, credible intervals

✓ "There is a 95% probability the effect is between A and B"

Nicenboim, B., & Vasishth, S. (2016). Statistical methods for linguistic research: Foundational ideas – part II. *Language and Linguistics Compass*, 10(8), 591–613. doi: 10.1111/lnc3.12207.
Vasishth, S., and Nicenboim, B. (2016) Statistical Methods for Linguistic Research: Foundational Ideas – Part I. Language and *Linguistics Compass*, 10(8), 349–369. doi: 10.1111/lnc3.12201.

The p-value: What It Is and Is Not

$$P(\text{data} \mid H_0 \text{ is true}) \neq P(H_0 \text{ is true} \mid \text{data})$$

✗ $p = .03$ means there's a 3% chance the null hypothesis is true	✓ It means if H_0 were true, we would see data this extreme only 3% of the time
✗ $p < .05$ means the effect is real or important	✓ Significance depends on sample size. With n = 10,000, tiny trivial effects become significant
✗ $p = .06$ means the effect does not exist	✓ It means the data are not sufficient to reject H_0 at $\alpha = .05$ (absence of evidence ≠ evidence of absence)
✗ A larger *p*-value means a smaller effect	✓ *p*-values conflate effect size and sample size; report effect sizes and CIs alongside *p*-values

Confidence Intervals vs. Credible Intervals

95% Confidence Interval (Frequentist)

Correct interpretation:
If we repeated this study 100 times, 95 of the intervals constructed this way would contain the true parameter.

What it does NOT mean:
There is a 95% probability the true value lies in this particular interval.

The true parameter is fixed; our value of interest is either in the interval or it is not.

95% Credible Interval (Bayesian)

Correct interpretation:
Given the data and our prior beliefs, there is a 95% probability the true parameter lies in this interval.

This is what most people think a CI is.

In R:
(`brms`, Bayesian Regression Models using 'Stan')

```
fit <- brm(RT ~ condition +
(1|participant) + (1|item), data = df)
posterior_interval(fit)
```

In practice, confidence intervals and credible intervals are often numerically similar.
However, the Bayesian interpretation is more intuitive and honest about uncertainty.

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Common Pitfalls

Some Pitfalls to Avoid

1

Ordinal data treated as continuous

→ Use clm() / clmm() for Likert ratings and visualize the distribution, not just the mean

2

Missing random effects

→ Always include (1|participant) and (1|item) at minimum; treatment of participants and items as independent inflates false positive rate

3

Overfit random effects structure

→ Maximal models often fail to converge; simplify with likelihood ratio tests or use Bayesian approaches (brms)

4

Uncorrected multiple comparisons

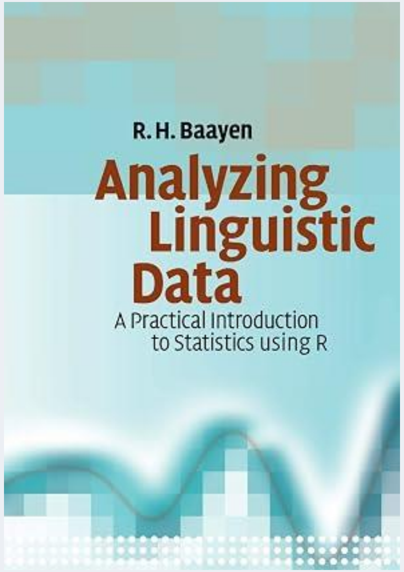
→ Bonferroni, false discovery rate (FDR) correction, and state your primary hypothesis explicitly before data collection

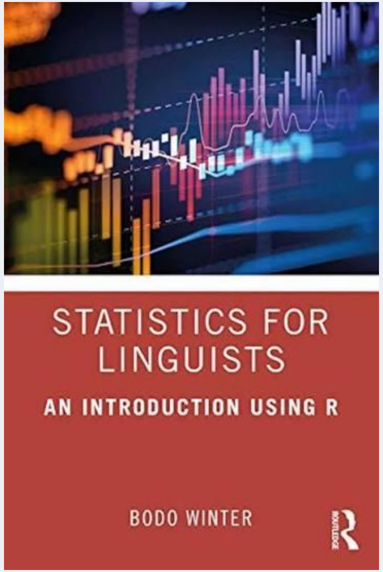
5

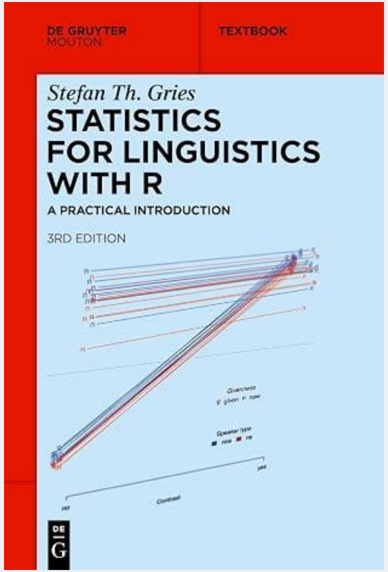
Confusing statistical and linguistic significance

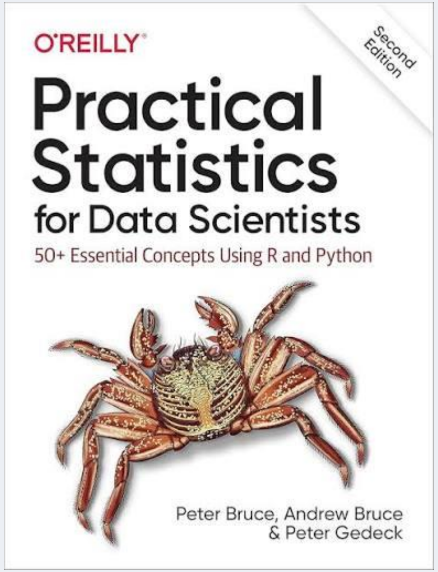
→ With large n, trivial effects are significant. Always report effect sizes (Cohen's d, partial R^2) alongside p-values

Useful Resources









Time to Practice

Google Colab

https://bit.ly/CLACCColab

(https://colab.research.google.com/drive/1DugdGaLlE_0XK4MQpOX5jLdtOZKPGsP4?usp=sharing)

GitHub

https://bit.ly/CLACCGitHub

(https://github.com/lhracs/CLA2026_crash_course)

✓ Synthetic study datasets available
✓ Code cells ready to run, no installation needed

Summary: Where are we now?

- The outcome variable type determines your analysis.
- Always visualize before you model.
- Include random effects for participants and items.
- A significant *p*-value is not the whole story.
- Statistics is a tool, but trust your linguistic knowledge.

Outcome type	Model
Continuous	`lmer()`
Binary	`glmer()`
Ordinal	`clmm()`
Count	`glmer(family = poisson)` `glmer.nb()`
Nominal	`multinom()`

What we covered

Variable types and data structure

The full analysis pipeline

Four model families with output interpretation

Model assumptions and how to check them

Frequentist vs. Bayesian framing

Common pitfalls

Where to go from here

Apply it: Practice! Use these slides as a reference for your next analysis

Stay critical: You now know enough to evaluate methods sections and spot when something is off

Thank you!

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